

ECEN-601: Homework 5

Due on October 28, 2025 at 11:59 pm

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Problem 1

Let V be a vector space over F . Show that the sum of two inner products on V is an inner product on V . Is the difference of two inner products an inner product? Show that a positive multiple of an inner product is an inner product.

Solution

We have two inner products $\langle \cdot, \cdot \rangle'$ and $\langle \cdot, \cdot \rangle''$ on V . A new inner product

$$\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle' + \langle \cdot, \cdot \rangle''$$

Since both inner products are inner products on V , the sum or difference of them retains the properties of linearity.

Additionally, conjugate symmetry is preserved:

$$\begin{aligned}\langle x, y \rangle &= \langle x, y \rangle' + \langle x, y \rangle'' \\ &= \overline{\langle y, x \rangle'} + \overline{\langle y, x \rangle''} \\ &= \overline{\langle y, x \rangle' + \langle y, x \rangle''} \\ &= \overline{\langle y, x \rangle}\end{aligned}$$

Problem 2

Let V be a real or complex vector space with an inner product. Show that the quadratic form determined by the inner product satisfies the *parallelogram law*

$$\|\alpha + \beta\|^2 + \|\alpha - \beta\|^2 = 2\|\alpha\|^2 + 2\|\beta\|^2.$$

Solution

We know $\|\mathbf{x}\|^2 = \langle \mathbf{x}, \mathbf{x} \rangle$. Thus,

$$\begin{aligned}\|\alpha + \beta\|^2 + \|\alpha - \beta\|^2 &= \langle \alpha + \beta, \alpha + \beta \rangle + \langle \alpha - \beta, \alpha - \beta \rangle \\ &= \langle \alpha, \alpha \rangle + \langle \alpha, \beta \rangle + \langle \beta, \alpha \rangle + \langle \beta, \beta \rangle + \langle \alpha, \alpha \rangle - \langle \alpha, \beta \rangle - \langle \beta, \alpha \rangle + \langle \beta, \beta \rangle \\ &= 2\langle \alpha, \alpha \rangle + 2\langle \beta, \beta \rangle \\ &= 2\|\alpha\|^2 + 2\|\beta\|^2 \quad (\text{by above mentioned definition})\end{aligned}$$

Problem 3

Show that the formula

$$\left\langle \sum_j a_j x^j \mid \sum_k b_k x^k \right\rangle = \sum_{j,k} \frac{a_j b_k}{j+k+1}$$

defines an inner product on the space $R[x]$ of polynomials with real coefficients. Let W be the subspace of polynomials of degree less than or equal to n . Find the weight matrix associated with this inner product on W , relative to the ordered basis $1, x, x^2, \dots, x^n$. (*Hint:* To show that the formula defines an inner product, observe that

$$\langle f \mid g \rangle = \int_0^1 f(t)g(t) dt$$

and work with the integral.)

Solution

If $f(t) = \sum_j a_j t^j$ and $g(t) = \sum_k b_k t^k$, then

$$\begin{aligned} \langle f \mid g \rangle &= \int_0^1 \left(\sum_j a_j t^j \right) \left(\sum_k b_k t^k \right) dt \\ &= \int_0^1 \sum_{j,k} a_j b_k t^{j+k} dt \\ &= \sum_{j,k} a_j b_k \int_0^1 t^{j+k} dt \\ &= \sum_{j,k} \frac{a_j b_k}{j+k+1} \end{aligned}$$

For a weight matrix G , $g_{ij} = \langle w_i, w_j \rangle$ where w_i is the i th basis vector. Thus, $g_{ij} = \sum_{k,l} \frac{a_k b_l}{k+l+1}$

Problem 4

Let V be the vector space of real polynomials with degree at most 3. Equip V with the inner product

$$\langle f | g \rangle = \int_0^1 f(t)g(t) dt.$$

- (a) Find the orthogonal complement of the subspace of constant polynomials.
- (b) Apply the Gram-Schmidt process to the basis $\{1, x, x^2, x^3\}$.

Solution

Part (a):

$$V^\perp = \{v \in V \mid \langle v, w \rangle = 0 \ \forall w \in W\}.$$

Part (b): Gram-Schmidt process:

$$\begin{aligned} u_1 &= 1 \\ u_2 &= x - \frac{\langle x, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 = x - \frac{\int_0^1 x \cdot 1 dt}{\int_0^1 1 \cdot 1 dt} \cdot 1 = x - \frac{1/2}{1} = x - \frac{1}{2} \\ u_3 &= x^2 - \frac{\langle x^2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle x^2, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 = \dots \end{aligned}$$

Problem 5

Let V be the vector space of all $n \times n$ matrices over C , with the inner product $(A|B) = \text{tr}(AB^H)$. Find the orthogonal complement of the subspace of diagonal matrices.

Solution

If V is the set of all $n \times n$ matrices over C , then the orthogonal complement of V is defined as

$$V^\perp = \{v \in V \mid \langle v, w \rangle = 0 \ \forall w \in W\}.$$

But now the inner product is given as $(A|B) = \text{tr}(AB^H)$. Thus,

$$V^\perp = \{A \in V \mid \text{tr}(AB^H) = 0 \ \forall B \in W\}.$$

Problem 6

Let W be a finite-dimensional subspace of an inner product space V , and let E be the orthogonal projection of V in W . Prove that $\langle E\alpha | \beta \rangle = \langle \alpha | E\beta \rangle$ for all α, β in V .

Solution

Scaling, by the definition of the inner product, we have

$$\langle E\alpha | \beta \rangle = E\langle \alpha | \beta \rangle$$

and since

$$\langle x, x' \rangle = \langle x', x \rangle$$

we have

$$\begin{aligned}\langle E\alpha | \beta \rangle &= E\langle \alpha | \beta \rangle \\ &= E\langle \beta | \alpha \rangle \\ &= \langle \alpha | E\beta \rangle\end{aligned}$$